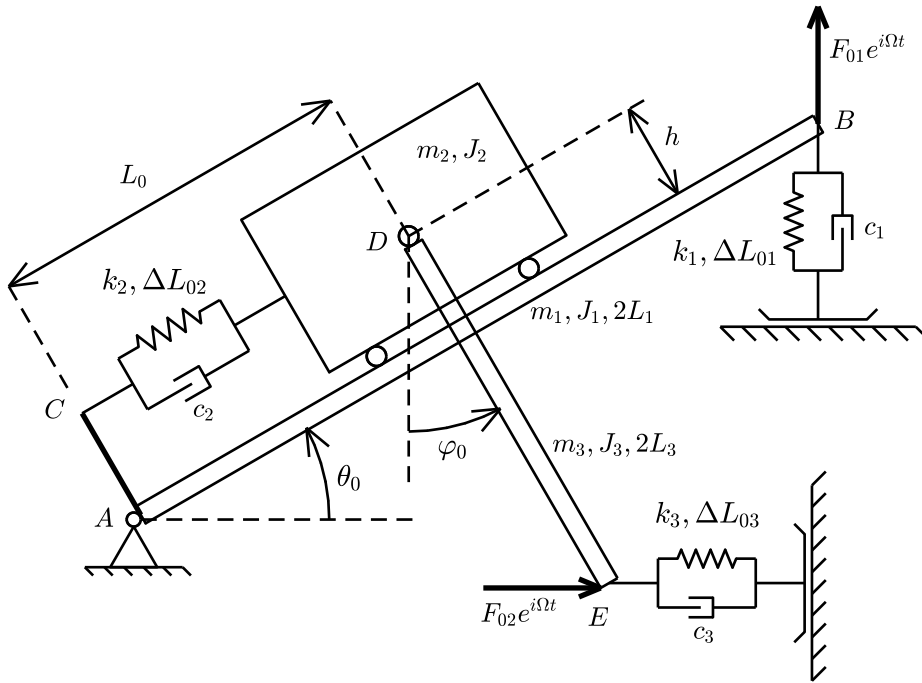


Dynamics of Mechanical Systems
September 05th, 2019
Standard problem (grading up to 30/30)



$m_1 = 10 \text{ kg}$	$\Delta L_{01} = -0.055 \text{ m}$
$m_2 = 5 \text{ kg}$	$\Delta L_{02} = -0.016 \text{ m}$
$m_3 = 10 \text{ kg}$	$\Delta L_{03} = 0.009 \text{ m}$
$J_1 = 0.2 \text{ kgm}^2$	$k_1 = 3.0\text{e}3 \text{ N/m}$
$J_2 = 0.1 \text{ kgm}^2$	$k_2 = 6.0\text{e}3 \text{ N/m}$
$J_3 = 0.2 \text{ kgm}^2$	$k_3 = 3.0\text{e}3 \text{ N/m}$
$L_1 = 0.5 \text{ m}$	$c_1 = 3 \text{ Ns/m}$
$L_3 = 0.5 \text{ m}$	$c_2 = 3 \text{ Ns/m}$
$L_0 = 0.75 \text{ m}$	$c_3 = 3 \text{ Ns/m}$
$h = 0.1 \text{ m}$	$\theta_0 = \pi/6$
	$\phi_0 = \pi/6$

The mechanical system above lays in the vertical plane and is shown in its equilibrium position.

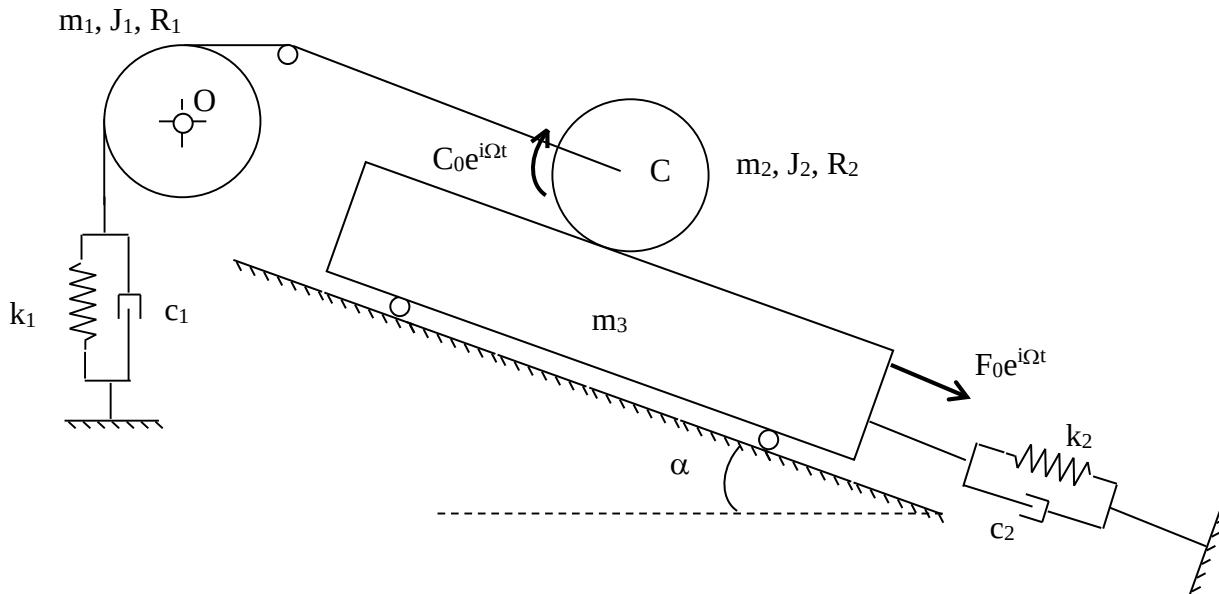
The students are requested to:

1. Write their surname, name, matriculation number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations. This answer shall be provided in written on paper;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the vertical harmonic force $F_{01}e^{i\Omega t}$. Output: the vertical component of the absolute displacement of the hinge D and the rotation of the bar DE. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic horizontal force $F_{02}e^{i\Omega t}$. Output: the horizontal component of the constraint force in the hinge A. Save the graphical result as SNxxx5.fig;
6. Compute the static preloads ΔL_{01} , ΔL_{02} , and ΔL_{03} for a static equilibrium position with $\theta_0 = \phi_0 = \pi/4$.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test standard September 05th 2019*
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Dynamics of Mechanical Systems
September 05th, 2019
Simplified problem (grading up to 22/30)



In the mechanical system above, shown in its equilibrium position and laid in the vertical plane, the disk of mass m_2 rolls without sliding on the cart m_3 .

The students are requested to:

1. Write their surname, name, matriculation number and SNxxx code (see note below) on the top of first page of each sheet, leaving blank field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;
2. Write the system's equations of motion around the given static configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the rotation of the disk m_1 and the displacement of the cart m_3 along the inclined plane. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the vertical component of the constraint force in the hinge O. Save the graphical result as SNxxx3.fig.

$m_1 = 45 \text{ kg}$	$R_1 = 0.3 \text{ m}$	$J_1 = 1.5 \text{ kgm}^2$	$k_1 = 30\text{e}3 \text{ N/m}$
$m_2 = 15 \text{ kg}$	$R_2 = 0.3 \text{ m}$	$J_2 = 0.5 \text{ kgm}^2$	$k_2 = 30\text{e}3 \text{ N/m}$
$m_3 = 40 \text{ kg}$	$\alpha = \pi/6$		$c_1 = 5 \text{ Ns/m}$
			$c_2 = 5 \text{ Ns/m}$

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test simplified September 05th 2019*
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.